Virtual Clinical Trial Design for Infective Endocarditis: Evaluating Antibiotic Therapy

Jacob W. Kammer, Klas Udekwu, and Esteban A. Hernandez-Vargas

Abstract

Methicillin-resistant *Staphylococcus aureus* (MRSA) endocarditis that arises from a bone infection relapses in 20% to 30% of patients despite appropriate antibiotic therapy. We developed a dual-compartment pharmacokinetic/pharmacodynamic model to examine which properties of the drug–host–reservoir system separate durable clearance of bacteremia from relapse. The model tracks vancomycin-sensitive and vancomycin-resistant MRSA in blood and in a bone reservoir that seeds the blood, and it simulates a clinical case of endocarditis secondary to ankle osteomyelitis: 4 days of vancomycin, 6 weeks of linezolid, and 14 days of follow-up. Populations falling below a limit of detection of $10\ \mathrm{CFU/mL}$ were treated as extinct. Linezolid potency ($EC_{50,L}$), linezolid maximum effect ($E_{max,L}$), and the immune clearance rate (k_{immune}) were each sampled from a log-normal distribution in separate Monte Carlo cohorts of 1000 simulated patients, and the value of each parameter at which the outcome changed was located with Brent's method. Vancomycin eliminated the sensitive strain, and linezolid cleared resistant MRSA from blood in nearly all patients, but the reservoir remained detectable in a minority and reseeded the blood after treatment ended. Relapse occurred in 22.0%, 22.6%, and 27.8% of the $EC_{50,L}$, $E_{max,L}$, and k_{immune} cohorts, almost always within two days of linezolid completion; persistent bacteremia occurred in at most 0.2%. The transition values were $0.76\ \mathrm{mg/L}$, $1.85\ \mathrm{h^{-1}}$, and $0.093\ \mathrm{h^{-1}}$, and the fraction of each distribution lying beyond its transition value (20.2%, 23.1%, and 27.9%) agreed with the simulated cohorts. Because the immune clearance rate and the parameter variabilities were calibrated to the published relapse range, these percentages are not independent predictions. The framework expresses relapse risk as the share of a patient population lying beyond a transition value, and it requires validation against measured blood and bone burdens.

I. Introduction

Infective endocarditis is a disease of the endocardial surface of the heart that involves the cardiac valves and the mural endothelium [1]. It is an indolent infection with Staphylococcus aureus (S. aureus) and Viridans streptococci as the most common pathogens [2]. It is an aggressive infection with an in hospital mortality rate of 15% to 20% and a one-year mortality rate

of 40% [3]. Complications include congestive heart failure, thromboembolic events, perivalvular and annular abscesses [4], sepsis and septic shock. Methicillin-resistant Staphylococcus aureus (MRSA) is the most common cause of bacterial endocarditis in much of the world [2].

Vancomycin has become the cornerstone of antibiotic treatment against MRSA infective endocarditis since the 1970s, and is the first line treatment against MRSA endocarditis [5]. Vancomycin, a glycopeptide, inhibits cell wall synthesis by preventing polymerization of glycan strands and cross linking between the strands via transpeptidase (PBP) [6]. Increased reliance on treatment with vancomycin since the 1980s has led to the development of resistant forms [7]. Infections of the heart values have become mixtures of MRSA sensitive and resistant to vancomycin. Compounding the problem of resistance is the challenge of treating the original source of infection, such as orthopedic hardware, central venous catheters and infection of the bone. It is well understood that S. aureus infections can be chronic and recurrent despite appropriate antibiotic treatment [8]. overarching all of these issues is the pivotal role of the immune response in resolving infections. Depending on the clinical scenario treatment failure rates can be as high as 20 to 30 percent. All of these factors in combination pose a significant clinical challenge that will require an improved understanding.

Since the end of the 20th century newer antibiotics, such as linezolid have been developed to target the antibiotic resistant S. aureus. Linezolid, an oxazolidinone antibiotic, has become one of the second line agents used in the treatment of vancomycin resistant MRSA endocarditis [9]. Beyond simply adding linezolid to treatment protocols an improved understanding of the factors that lead to clinical resolution (bacteremia resolved) or failure (persistent or relapsing bacteremia) is needed. Understanding how bacterial phenotypic variability, antibiotic combinations, and host factors interact will lead to improved outcomes in bacterial endocarditis.

Mikkaichi and colleagues [10] developed a mathematical model to examine the interplay of phenotypic heterogeneity with host factors in the persistence of MRSA bacteremia. Mikkaichi used ensemble modeling and other ML techniques to determine the key differences between resolving and persistent bacteremia [10]. Two phenotypically distinct bacterial populations were used, one phenotypically normal, the other composed of persister cells. Persister cells are defined as slower growing and are less susceptible to antibiotics [10]. Mikkaichi's model analyzed the interaction of vancomycin, host immune function and differences in bacterial phenotypic behavior to identify the key factors that lead to persistent bacteremia. Using this ensemble modeling approach, Mikkaichi and colleagues identified decreased immune clearance rate of persister

cells as the primary determinant of persistent bacteremia [10].

Vera-Yunca et al. developed a two-state semi-mechanistic PKPD model comprising a growing, drug-susceptible population and a non-growing, drug-insusceptible dormant state. The model was built on in vitro time–kill experiments with linezolid and vancomycin against MRSA and subsequently adjusted to data from a neutropenic mouse pneumonia model. The in vivo drug-effect parameters were obtained by re-estimating scaling ratios relative to the in vitro estimates rather than independently. Whereas Mikkaichi et al. incorporate host immune clearance explicitly, the neutropenic design of Vera-Yunca's model excluded it.

The present study uses a dual-compartment mathematical Pharmacokinetic/pharmacodynamic model (PKPD) to examine the interplay of the factors that lead to resolution of bacteremia in the setting of osteomyelitis and bacterial endocarditis. This type of model is used to connect drug exposure to drug response. Linking time course of drug concentration to the time course of drug effect allows the model to simulate what happens during treatment. Two MRSA populations are used across the two compartments. One MRSA subpopulation is vancomycin resistant while the other is sensitive to vancomycin. The host system is characterized by a clinical diagnosis of bacterial endocarditis with the initial bacteremia secondary to osteomyelitis of the right ankle. The model evaluates how these two phenotypically distant MRSA sub populations grow in the environment of antibiotic treatment, and immune response of in-silico patients. Using well understood population averages of clinical relapse in a specific clinical scenario, the techniques of Monte carlo sampling and non-linear dynamics are used to delineate what properties of the drug–host–reservoir system produce durable bacteremia clearance or post-treatment relapse. By integrating bacterial dynamics, antibiotic pharmacology, and the host innate immune response within a single modeling framework, this approach provides a means of studying the interplay between these forces in a way that human clinical trials cannot.

II. Methods

A. Case Study: Clinical Scenario

Bacterial endocarditis, an infection of the heart valves, frequently develops as a complication of a deep, unresolved bone infection known as osteomyelitis. In this scenario, bacteria from the infected bone continuously leak into the bloodstream creating a bacteremia. The bacteremia allows bacteria to travel to the heart and attach to its valve structures. The following case

illustrates this clinical pathway and serves as the foundation for the proposed pharmacokinetic and pharmacodynamic simulation study.

The case involves a 62-year-old man with type 2 diabetes who was admitted to the hospital after outpatient antibiotic treatment failed to clear a chronic, non-healing wound on his right ankle. Because diabetes impairs both circulation and immune function, the infection had spread deep into the bone, causing right ankle osteomyelitis. Upon admission, the patient was ill, presenting with progressive fatigue and high, intermittent fevers peaking at 39°C. Physical examination showed a wound on the right ankle draining clear blood, while laboratory work revealed elevated C-reactive protein (a marker of severe systemic inflammation) and a high white blood cell count. Although a physical examination of his heart was unremarkable, a transesophageal echocardiogram (TEE), a specialized ultrasound probe passed down the esophagus to view the heart up close—showed thickening of the mitral valve's posterior leaflet, confirming the diagnosis of bacterial endocarditis.

Based on a presumptive diagnosis of staphylococcal endocarditis, blood cultures were drawn, and the patient was started on intravenous vancomycin at 1,000 mg every 8 hours. Despite this treatment, he continued to spike fevers. On the third day of hospitalization, blood culture results identified the organism as methicillin-resistant *Staphylococcus aureus* (MRSA) with a vancomycin Minimum Inhibitory Concentration (MIC) of 32 mg/L. This high MIC indicated that the strain was resistant to vancomycin, explaining why the patient was not improving. Laboratory testing showed that the bacteria were sensitive to linezolid. Vancomycin was discontinued, and the patient was switched to linezolid at 600 mg every 12 hours. Over the next 48 hours, his fever curve normalized.

To complete the treatment for MRSA endocarditis, the patient required a 6-week course of linezolid in accordance with Infectious Diseases Society of America (IDSA) guidelines. The primary clinical endpoint of this therapy is the complete resolution of bacteremia through the eradication of the primary infection source, which in this patient is the right ankle osteomyelitis. However, even with effective antibiotic therapy, 20% to 30% of patients experience a relapse of bacteremia because the infected bone continues to seed bacteria into the blood unless surgical intervention is performed to remove the source. The proposed dual-compartment pharmacokinetic/pharmacodynamic model simulates this exact dynamic. It tracks how linezolid distributes through body tissues (pharmacokinetics) and suppresses bacterial growth (pharmacodynamics) in the presence of a persistent, seeding bone infection.

B. Bacterial System and ODE Model

The dynamical framework consists of four state variables, S, R, S^r, and R^r, each representing the four distinct subpopulation of MRSA in the two compartments. Two of the subpopulation reside in the blood, vancomycin sensitive, S and resistant MRSA, R and the other two represent vancomycin sensitive and resistant MRSA in the reservoir, S^r and R^r. All four subpopulations are governed by a coupled system of four ordinary differential equations. Equations (1) and (2) relate to processes in the vascular compartment, Equations (3) and (4) relate to processes in the reservoir compartment. The natural death rate of bacteria was omitted from the model due to it's small size compared to the other variables.

$$\frac{dS}{dt} = \rho_s\, S\left(1 - \frac{S+R}{K}\right) - k_{\text{imm}}\, S - E^V(C_V)\, S - E^L(C_L)\, S + k_{r\to b}\, S^r. \tag{1}$$

$$\frac{dR}{dt} = \rho_r\, R\left(1 - \frac{S+R}{K}\right) - k_{\text{imm}}\, R - E^L(C_L)\, R + k_{r\to b}\, R^r. \tag{2}$$

$$\frac{dS^r}{dt} = \rho_s^r S^r\left(1 - \frac{S^r+R^r}{K^r}\right) - 0.5\, k_{\text{imm}}\, S^r - E^V(\phi_V C_V)\, S^r - E^{L,r}(\phi_L C_L)\, S^r - k_{r\to b}\, S^r. \tag{3}$$

$$\frac{dR^r}{dt} = \rho_r^r R^r\left(1 - \frac{S^r+R^r}{K^r}\right) - 0.5\, k_{\text{imm}}\, R^r - E^{L,r}(\phi_L C_L)\, R^r - k_{r\to b}\, R^r. \tag{4}$$

Each drug effect term is a sigmoidal E_{max} relationship with a Hill coefficient of unity, evaluated at the free-drug concentration reaching that compartment:

$$E^i(C) = \frac{E^i_{\max}\, C}{EC^i_{50} + C}, \qquad i \in \{V, L\}, \tag{5}$$

where C_V and C_L are the free vancomycin and linezolid concentrations in blood from Eq. (7), and the reservoir sees only the penetrated fractions $\phi_V C_V$ and $\phi_L C_L$. The maximum rate of linezolid killing in the reservoir is reduced by an independent efficacy factor, $E^{L,r}_{\max} = \alpha\, E^L_{\max}$ with $\alpha = 0.45$. This term reflects the reduced activity of linezolid against slow-growing, biofilm-embedded and small-colony-variant *S. aureus* in bone. The factor α is seperate from the reservoir growth rates and is an assumed value, estimated by calibration. It gives an effective reservoir maximum linezolid killing rate of $0.45 \times 1.97 \approx 0.89\ \text{h}^{-1}$. Vera-Yunca et al. estimated that linezolid $E_{\max}$ in a neutropenic mouse pneumonia model was 0.303 times its in vitro value (95% CI 0.269–0.330), an in vivo maximum of approximately $0.60\ \text{h}^{-1}$. and attributed the reduction to

factors other than drug exposure at the infection site [11]. Because the present model represents reduced bone exposure explicitly through the penetration fraction ϕ_L, part of that attenuation is already accounted for, and α is correspondingly larger than 0.303; it lies within the 0.2–0.6 range over which that attenuation is uncertain. $E^{L,r}$ denotes the resulting effect term.

Both compartments have distinct logistic growth dynamics and the carrying capacities of each compartment is shared between the sensitive and resistant species. The logistic term depends on the combined density $S+R$ (blood) or S^r+R^r (reservoir), so the strains compete directly for the same niche. Vancomycin acts on the sensitive strain only. The immune response in the reservoir is scaled to a fraction of the blood compartment. As detailed previously, the reservoir acts as a seeding source for both sensitive and resistant populations entering the blood compartment. Return movement from blood to reservoir is neglected.

The system of ODEs was integrated numerically in Python with SciPy's `solve_ivp` function using the LSODA method, a relative tolerance of 10^{-6}, an absolute tolerance of 10^{-8} $\mathrm{CFU/mL}$, and a maximum step of 4 h [12].

C. Model Calibration

The initial condition of the reservoir assumes inoculum of 100 for sensitive and resistant bacterial species respectively. The blood compartment at the beginning of the simulation is assumed to be free of pathological bacterial species $S(0) = 0$ and $R(0) = 0$, which is consistent with normal human physiology. Values for vancomycin and linezolid EC_{50} and E_{max} reported in the literature are variable and likely due to methodological differences between studies. This limitation makes it more challenging to use all of the limited amount of data currently available.

To provide internal consistency, growth rates and pharmacodynamic values (E_{max} and EC_{50}) for both vancomycin and linezolid were taken from Vera-Yunca and colleagues [11]. The study used in vitro time-kill and in vivo neutropenic mouse lung data to fit growth rates and antibiotic PD parameters. The in-vitro data was fit to the mouse data with a scaling factor. In our study we used the in-vitro data to represent the blood compartment and the in-vivo, the reservoir compartment. We used a scaling factor to adjust the growth rate of vancomycin sensitive MRSA in the reservoir and linezolid's $E_{max,L}$. No changes were made to the $EC_{50,L}$. The scaling factor used was based on calibration and not the same value used by Vera-Yunca and colleagues. The percentage of simulated cases where bacteremia is present by the end of the followup period are based on population averages [13].

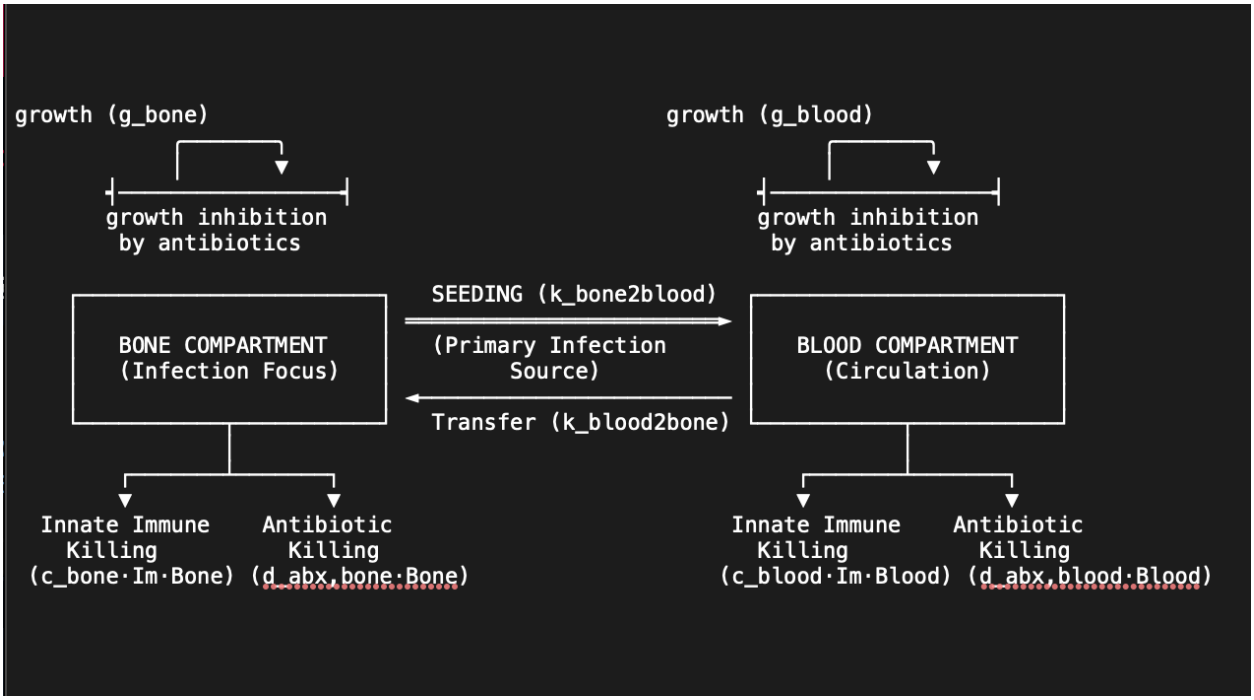


Fig. 1. Two Compartment Model

D. Blood Compartment

The blood compartment represents the circulatory system of an individual with blood moving rapidly between the organs and throughout the body. There is no attenuation of bacterial killing by the immune system. Carrying capacity is shared between sensitive and resistant bacteria and represents the maximum bacterial load that can be sustained within the vascular compartment. The carrying capacity is based on literature estimates from [14]. The growth rate of sensitive SA is set to 1.47 h^{-1}, the in vitro estimate for MRSA reported by Vera-Yunca and colleagues [11], treating well-mixed, nutrient-replete blood as in-vitro-like. This value corresponds to a doubling time of 28 minutes. The resistant SA growth rate is scaled by a 10% fitness cost to 1.323 h^{-1}, doubling time of 31 minutes. The microbiological parameters are detailed in table I.

E. Reservoir Compartment

The reservoir compartment models the osseous environment and it the initial source of infection after having spread from the ankle wound to the long bone. Infected bone undergoes localized inflammation and creates sequestrum (necrotic bone surrounded by infected granulation tissue) with poor immune cell access and restricted antibiotic diffusion [15]. Staph aureus forms biofilms

that adhere to the sequestrum forming an organized structure with the bacteria surrounded by a matrix. Biofilm can release planktonic microbes from their surface and seed the circulation, eventually leading to new biofilms at other locations such as the mitral valve. Staph aureus forms Small Colony variants (SVC) that can reside inside osteoblasts, providing further protection from antibiotics and the immune system[8]. SCVs are slow-growing phenotypic variants that can form colonies and have been isolated from infections such as osteomyelitis and endocarditis [8]. These factors lead to slower growth and lower metabolic activity in bone driving a state of chronic persistence. The mathematical model takes these physiological and microbiological characteristics into consideration by assuming lower growth rates between the two compartments. The growth rate for sensitive bacteria is 0.845 h^{-1} in the reservoir, corresponding to a doubling time of 49 minutes; the resistant strain carries the same 10% fitness cost as in blood (0.761 h^{-1}, 55 minutes). The lack of data on bone growth rates requires the use of a surrgoate value to represent bone. The reservoir value is the in vivo maximum growth rate that Vera-Yunca and colleagues estimated in a mouse lung model, 42% lower than their in vitro rate of 1.47 h^{-1} [11]. Its use for bone assumes that the in vivo reduction in growth applies similarly to poorly perfused, biofilm-associated bone. Exchange between compartments is assumed to be one-way and governed by a first-order rate constant $k_{r\to b}$ (reservoir to blood). Return movement of bacteria from blood into the reservoir is negligible compared with shedding from the established bone focus into the blood. This is an assumed value due to the lack of human experimental measurements. Similarly to the blood compartment, the reservoir carry capacity is shared between sensitive and resistant bacteria, however, unlike the blood compartment it is assumed to be higher. This assumption is based on the assumed presence of greater nutrient sources in necrotic bone and values reported in the literature [11].

$$R^b + S^b \xleftarrow{k_{r\to b}} R^r + S^r \tag{6}$$

TABLE I

COMPARTMENTAL PARAMETER VALUES

Parameter	Units	Nominal Value	Reference
ρ_s^b	h^{-1}	1.47	[11]
ρ_r^b	h^{-1}	1.323	10% fitness cost
ρ_s^r	h^{-1}	0.845	[11]
ρ_r^r	h^{-1}	0.761	10% fitness cost
$B_{\max}^b$	CFU/mL	6×10^3	[14]
$B_{\max}^r$	CFU/mL	1×10^7	[11]
$k_{r \to b}$	h^{-1}	1×10^{-7}	[16]

F. Immune Response

The onset of infection activates the body's innate immune system. White blood cells (WBCs), particularly neutrophils, which are largely quiescent prior to infection, become activated and migrate to sites of bacterial invasion. Once at the site of infection, these cells release bactericidal substances, including reactive oxygen species and proteolytic enzymes, before undergoing cell death. Pro-inflammatory cytokines and chemokines released at the site of infection stimulate the bone marrow to increase WBC production and mobilization, leading to rapid delivery of immune cells through the circulatory system to infected tissues.

Host immune clearance is represented as a constant, first-order killing term. All four bacterial subpopulations in both blood and reservoir are cleared at a rate proportional to their own density and to a single rate constant, k_{immune} (baseline $0.18\ \mathrm{h}^{-1}$). The reservoir immune response is scaled to 0.5 of the response in blood. This reduction models biofilm exclusion of immune cells and the intracellular protected state of MRSA. The model for immune response is an assumed

and simple representation of a highly complex biological process that depends on a multitude of factors.

G. Antibiotic Pharmacokinetics

A standard protocol for the treatment of infectious endocarditis is used, based on the recommendations of the Infectious Disease Society of America ([5]). Vancomycin is the first line agent and linezolid secondarily once vancomycin resistance was been recognized. Vancomycin, a weakly bactericidal agent, inhibits bacterial cell wall synthesis and linezolid, a bacteriostatic agent, inhibits protein synthesis. Studies have shown that linezolid possesses both intracellular and extracellular activity, against Staphylococcus[17]. During simulation, guideline doses, Dose_i, and dosage time intervals, τ_i, for vancomycin and linezolid are used [18]. Total drug administration time is parameterized by Duration.

Serum drug concentrations are simulated with a one-compartment PK model for both Vancomycin and Linezolid. Vancomycin is approximately 50% protein-bound in plasma, whereas linezolid is more modestly protein-bound ($\sim$30%, i.e., $\sim$70% free fraction), a difference the model accounts for via each drug's free-fraction parameter (f_u = 0.50 for vancomycin, 0.70 for linezolid) when converting total plasma concentration to C_i, the active free-drug concentration. Antibiotic clearance, CL_i is consistent with normal kidney function. Population averages are used for the apparent volume (liters) of distribution, V_d.

Between doses, Dose_i, the concentration decays via first-order kinetics with elimination rate $k_{e,i} = CL_i/V_{d,i}$. At each dosing time $t_k = k\,\tau_i$ the concentration steps up by $\mathrm{Dose}_i/V_{d,i}$ and then decays again, so that within the k-th dosing interval

$$C_i(t) = C_i(t_k^+)\, e^{-k_{e,i}(t-t_k)}, \qquad t_k \le t < t_{k+1}, \tag{7}$$

with the post-dose value set by the accumulation rule

$$C_i(t_k^+) = C_i(t_k^-) + \frac{\mathrm{Dose}_i}{V_{d,i}}, \qquad C_i(0^-) = 0, \tag{8}$$

for $i \in \{V, L\}$ (vancomycin, linezolid). Each drug is dosed only within its own course window. Outside the window no further doses are added and the concentration decays toward zero. Multiplying by the unbound fraction $f_{u,i}$ gives the free-drug concentration that drives pharmacological effect.

TABLE II

TABLE OF PHARMACOKINETIC PARAMETER VALUES

Parameter	Units	Nominal Value	Reference
$E^V_{\max}$	h^{-1}	1.7	[11]
$E^L_{\max}$	h^{-1}	1.97	[11]
EC^V_{50}	mg/L	0.25	[11]
EC^L_{50}	mg/L	0.56	[11]
V^V_d	Liters	50	[19]
V^L_d	Liters	45	[20]
CL_V	ml/min	83 (5.0 L/h)	[19]
CL_L	ml/min	100 (6.0 L/h)	[20]
ϕ_{LZD}	–	0.50	[20] [21]
ϕ_{VAN}	–	0.15	[15] [22]
α	–	0.45	[11]

There is significant variability reported in the literature on the degree of bone penetration of linezolid, although clinicians generally view it's bone penetration as favorable and significantly greater than vancomycin. Linezolid bone penetration was set to 50% creating free linezolid concentration from 1.2 to 5.9 mg/L, averaging 2.9 mg/L. This estimate compares favorably to data reported by [21] using the same dosing regimen. That study reported a mean bone concentration of 3.9 mg/L (range: 1.2 - 8.3 mg/L). Vancomycin bone penetration was set to 15%.

During simulation the steady state peak Vancomycin concentration reached 36.3 mg/L and a

trough concentration of 16.3 mg/L. The total-drug area under the concentration-time curve over 24 hours of $\mathrm{AUC}_{24,\mathrm{ss}} = 600\ \mathrm{mg \cdot h/L}$, independent of volume of distribution or half-life. This falls at the upper edge of the standard clinical target of 400–600 mg·h/L for MRSA bacteremia and endocarditis[23]. Accounting for the drug's free fraction ($f_u = 0.50$), the corresponding free-drug exposure driving bacterial kill in the model is $\mathrm{AUC}^{\mathrm{free}}_{24,\mathrm{ss}} = 300\ \mathrm{mg \cdot h/L}$. Linezolid steady-state total-drug trough concentration was calculated at 3.4 mg/L, which is within the recommended range of 2 to 8 mg/L if the MIC is less than 2 mg/L[24].

H. Definition of treatment Failure and Success

Treatment failure is defined as the presence of bacteremia at the end of the simulation and success as resolution of the bacteremia. The simulation ends at the conclusion of the antibiotic free period. When the concentration of MRSA drops below the Level of Detection, $10\ \mathrm{CFU/mL}$ the model resets the concentration to zero. The limit of detection (LOD) functions as an extinction boundary where any bacterial subpopulation that fell below the LOD are set to zero. MRSA cannot grow back from a fractional leftover remnant. Without such a boundary, a deterministic ODE model allows a population far below one organism to persist and later regrow, an artifact of representing a discrete population with continuous variables [25], [26]. The value assigned to the limit of detection is a modeling assumption. An extinct blood subpopulation in blood can be re-established if continued transfer from a detectable reservoir accumulates to the limit of detection (LOD).

Treatment failure is evaluated by two separate and mutually exclusive endpoints. Let $R_B(t)$ denote the MRSA density in blood at time t (days) and t_L the time of linezolid completion. A persistent bacteremia is defined as detectable resistant MRSA in blood at the end of linezolid treatment,

$$\mathcal{P} = \Big\{ R_B(t_L) \geq \mathrm{LOD} \Big\}. \tag{9}$$

Relapse of the bacteremia is defined as clearance of resistant MRSA from blood at linezolid completion followed by a return to detectable levels during the 14-day follow-up,

$$\mathcal{R} = \Big\{ R_B(t_L) < \mathrm{LOD} \Big\} \cap \Big\{ \exists\, t \in (t_L,\, t_L + 14] : R_B(t) \geq \mathrm{LOD} \Big\}. \tag{10}$$

These two conditions at t_L and the simulation results are reported separately for each outcome. Separate Monte Carlo simulations are performed for each of the 3 parameters over the experimental period of 81 days. This produces a spread of possible outcomes (trajectories) for each

of the four groups of MRSA. It is assumed that the relapsing bacteremia is due to a reseeding of the blood compartment from the reservoir.

I. Event location at the limit of detection

Extinction and reactivation at the LOD are handled inside each simulation by event location. This procedure acts on time, within a single simulation. It is separate from the determination of the transition points described in Section II-J, which acts on a parameter, across simulations.

The extinction boundary as represented by the limit of detection was enforced with the event detection of `solve_ivp` [12]. Each of the four subpopulations y_i was assigned an event function

$$g_i(t) = y_i(t) - \mathrm{LOD}. \tag{11}$$

A detectable subpopulation was monitored for a downward crossing (extinction) and an extinct subpopulation for an upward crossing (reactivation). The integrator compared the sign of g_i at the two ends of every integration step. When the sign changed, it located the crossing time within that step by applying Brent's method to g_i, evaluated with the integrator's continuous interpolant, to a tolerance of four units of machine precision [27], [28]. Because g_i is continuous in time and changes sign over the step, a true root exists.

At every event the integration was stopped at the crossing time. The subpopulation was set to exactly zero or restarted at the LOD, and the integration was resumed from that time. The trajectory was computed as a sequence of continuous segments separated by discrete resets [29], [30]. While extinct, a subpopulation had no growth, antibiotic killing, immune clearance, or outgoing transfer. Only incoming transfer from a detectable subpopulation in the reservoir into the blood compartment was accumulated. Dose administration times were not treated as events and were resolved by the $4\ \mathrm{h}$ maximum step.

Event location therefore determines when, within one simulation, each subpopulation becomes extinct or is re-established. It does not determine the value of any parameter.

J. Determination of transition points with Brent's method

Each of the parameters $EC_{50,L}$, $E_{\max,L}$, and k_{immune} has a transition value that separates two outcomes at the end of the simulation: clearance of the infection or its persistence. These transition values were located using the Brent algorithm. Let θ denote the parameter being varied, with all other parameters held at baseline, and let $R_{res}(t_f;\theta)$ denote the resistant MRSA burden

in the reservoir at the final simulated time t_f. One complete 81-day ODE simulation was run for each evaluation of

$$F(\theta) = R_{res}(t_f; \theta) - \mathrm{LOD}, \qquad \mathrm{LOD} = 10 \ \mathrm{CFU/mL}, \tag{12}$$

so that $F(\theta) < 0$ denotes a reservoir that has been cleared at t_f and $F(\theta) > 0$ a reservoir that remains detectable. F is defined on the reservoir and not on the blood. It nevertheless separates resolution of the bacteremia from no resolution, because a detectable reservoir reseeds the blood once linezolid is stopped: at every grid point evaluated, the resistant blood burden at t_f was detectable exactly when the reservoir burden was.

Grids spanning $[0.05,\ 3.3]$ mg/L for $EC_{50,L}$, $[0.70,\ 2.2]$ h^{-1} for $E_{\max,L}$, and $[0.02,\ 0.65]$ h^{-1} for k_{immune} were evaluated first to verify that the sign of F changes exactly once across each range. The two adjacent grid points on opposite sides of that change supplied a sign-changing bracket $[a, b]$ with

$$F(a)\, F(b) < 0. \tag{13}$$

SciPy's `scipy.optimize.brentq` implementation was then applied to $F(\theta)$ on this bracket [12]. The method combines bisection, secant interpolation, and inverse quadratic interpolation. An interpolated step is accepted only when it falls inside the current bracket and shortens the step sufficiently; otherwise a bisection step is taken. After each evaluation the new point replaces the endpoint at which F has the same sign, so the bracket always satisfies Eq. (13) and always encloses the transition. The search stops when the width of the bracket falls below the tolerance, which was set to 10^{-3} for $EC_{50,L}$ (mg/L) and $E_{\max,L}$ (h^{-1}) and to 10^{-6} h^{-1} for k_{immune}. The reported value therefore lies within the tolerance of the transition. Transition values are reported to two significant figures, except for $E_{max,L}$, reported to three because its narrow sampling distribution makes the difference between 1.85 and 1.9 material.

The determination of a transition point is distinct from the event location of Section II-I; the two are compared in Table III. Event location finds a time within one simulation and is followed by a change of the equations and a restart of the integration. The transition search finds a parameter value across simulations: each evaluation of F is one complete simulation, including all of its event locations, and nothing is switched or restarted. The two procedures share only the root-finding algorithm. `solve_ivp` applies Brent's method to g_i internally, and the same algorithm was applied here to F. They differ in what that algorithm can deliver: g_i is continuous in time, so a root is found, whereas F is discontinuous in θ, so the search localizes

a change of outcome and not a root. The two are connected in one direction only. The reset to zero at the LOD within each simulation is what makes F discontinuous.

	Event location	Transition point
Scope	Within one simulation	Across simulations
Unknown	Time of an LOD crossing	Parameter value θ
Function	$g_i(t)$, Eq. (11)	$F(\theta)$, Eq. (12)
Continuity	Continuous in t	Discontinuous in θ
Result	A root, $g_i = 0$	Location of a change of sign
Action that follows	Equations switched, integration restarted	None; the value is reported

TABLE III

EVENT LOCATION AT THE LIMIT OF DETECTION COMPARED WITH THE DETERMINATION OF A TRANSITION POINT. BOTH USE BRENT'S METHOD, BUT FOR DIFFERENT PURPOSES.

F is discontinuous because of the hard LOD rule. A reservoir population that falls below the LOD is set to zero and cannot regrow, so on the clearance side $R_{res}(t_f;\theta) = 0$ and $F(\theta) = -\mathrm{LOD}$ exactly. On the persistence side the reservoir has regrown toward its carrying capacity by t_f and $F(\theta)$ is of order 10^6 to 10^7 $\mathrm{CFU/mL}$. F therefore jumps between these two levels at the transition and takes no value in between; in particular, there is no θ in the bracket with $F(\theta) = 0$. For $EC_{50,L}$, simulations immediately below and above the located transition produced terminal burdens (R_b, R_{res}) of $(0, 0)$ and approximately $(5.2 \times 10^3,\ 8.8 \times 10^6)$ $\mathrm{CFU/mL}$, respectively.

Brent's method remains applicable in this setting. Its termination does not depend on continuity: the bisection safeguard guarantees that, for any function satisfying Eq. (13), the bracket is reduced to the requested width while F keeps opposite signs at its two endpoints [31], [32]. Continuity is required only for the further conclusion that the final bracket contains a zero of the function, which is why it is listed as an assumption of `brentq` [33]. For a discontinuous F the final bracket instead contains the point at which the sign changes. The returned value is accordingly interpreted as the location of the change of outcome, to within the tolerance, and not as a parameter satisfying $F(\theta) = 0$. Persistence ($F > 0$) occurs above the transition value for $EC_{50,L}$ and below it for $E_{\max,L}$ and k_{immune}.

K. Probability of Bacteremia Clearance

In the prior section the Brent method is used to calculate the boundary point that separates the two clinical outcomes. The values of each parameter are modeled with the log normal distribution, which clusters most values around the mean with a small number of parameter values spread far above in the tail [34]. For the log-normal $EC_{50,L}$, $E_{max,L}$, and k_{immune} distributions, arithmetic mean m and coefficient of variation CV are converted to log-space parameters as

$$\sigma_{\ln} = \sqrt{\ln(1 + CV^2)}, \qquad \mu_{\ln} = \ln(m) - \frac{\sigma_{\ln}^2}{2}. \tag{14}$$

The distance between the boundary and the specific parameter value is calculated in unit of standard deviation. In the example of drug potency, $EC_{50,L}$ the bacteremia does not resolve by the end of the simulation if the $EC_{50,L}$ is located significantly above the boundary. For the other two parameters, maximum drug effect and immune response the bacteremia is not resolved by the termination of the simulation if the values lie significantly below the boundary. Each boundary point T, is standardized log-space distance from the specific parameter distribution is calculated by

$$z_T = \frac{\ln(T) - \mu_{\ln}}{\sigma_{\ln}}. \tag{15}$$

The symbol Φ denotes the standard-normal cumulative distribution function and represents the probability that bacteremia does not resolve by the end of the simulation is

$$p_{bacteremia} = 1 - \Phi(z_T) \quad \text{for } EC_{50,L}, \qquad p_{bacteremia} = \Phi(z_T) \quad \text{for } E_{max,L} \text{ and } k_{immune}, \tag{16}$$

Changes in the distribution used to model the parameter value will alter the distance and location of the parameter in relation to the boundary point. Change in the location of the probability mass with respect to the boundary will affect whether the bacteremia has resolved by the end of the simulation.

III. Results

A. Pre-Treatment Period

Prior to initiation of vancomycin, bacterial populations developed in the reservoir and seeded the blood compartment. The vancomycin-sensitive *S. aureus* strain in blood had growth rate $\rho_s = 1.47$ h^{-1} while the resistant strain had $\rho_r = 1.323$ h^{-1}, a 10% fitness cost. In the bone reservoir, replication rates were reduced to $\rho_s^r = 0.845$ h^{-1} for the sensitive strain and $\rho_r^r = 0.761$ h^{-1} for the resistant strain, the same 10% cost. At the start of vancomycin the baseline reservoir held

about 8×10^6 CFU/mL of the sensitive strain and 3×10^4 CFU/mL of the resistant strain. Once vancomycin has eradicated the senstive strain the resistant population grows up to the carrying capacity of the compartment.

The simulation represents progression from osteomyelitis, with associated wound of the ankle, to secondary bloodstream infection and subsequent development of bacterial endocarditis. Vancomycin administration (1000 mg every 8 hours) models clinically recommended steady-state peak and trough plasma concentrations and drives rapid killing of susceptible *S. aureus* subpopulations (S_b and S_{res}). Vancomycin exerted no growth inhibition on resistant strains (R_b and R_{res}).

B. Vancomycin Treatment Period

The 4-day vancomycin course ($\texttt{van_duration} = 96$ h, starting at $\texttt{vanco_start} = 504$ h) drives the vancomycin-sensitive strain to extinction in both blood (S_b) and reservoir (S_{res}) compartments before the initiation of linezolid. Following laboratory confirmation of vancomycin-resistant MRSA, therapy was switched to Linezolid (600 mg every 12 hours), yielding recommended steady-state serum concentrations. Linezolid administration drove a reduction in resistant bacterial counts across both compartments. However, clearance from the reservoir lagged significantly behind vascular clearance.

C. Linezolid Potency

To assess how variation in linezolid potency affects predicted *Staph aureus* bacterial kinetics, $EC_{50,L}$ was sampled from a log-normal distribution with mean $0.56\ \mathrm{mg/L}$ and coefficient of variation $\mathrm{CV} = 0.60$ [34]. The mean was taken from the linezolid pharmacodynamic estimate of Vera-Yunca et al. ($0.56\ \mathrm{mg/L}$) [11]. All other parameters were held at baseline. Each of the $n = 1000$ independently sampled $EC_{50,L}$ values were propagated through the model from blood-sterile, reservoir-seeded initial conditions ($S_b(0)$=$R_b(0)$=0, $S_{res}(0)$=$R_{res}(0)$=100 CFU/mL) to create a distribution of simulated trajectories fig. 2. Sensitive SA in both compartments is eradicated by vancomycin by the end of it's treatment course. The majority of the simulations the vancomycin resistant SA are eradicated in the blood compartment, but persist, above the level of detection in the reservoir compartment. Termination of linezolid treatment allows the reservoir to re-seed the blood compartment causing the relapse of the bacteremia during the follow up period.

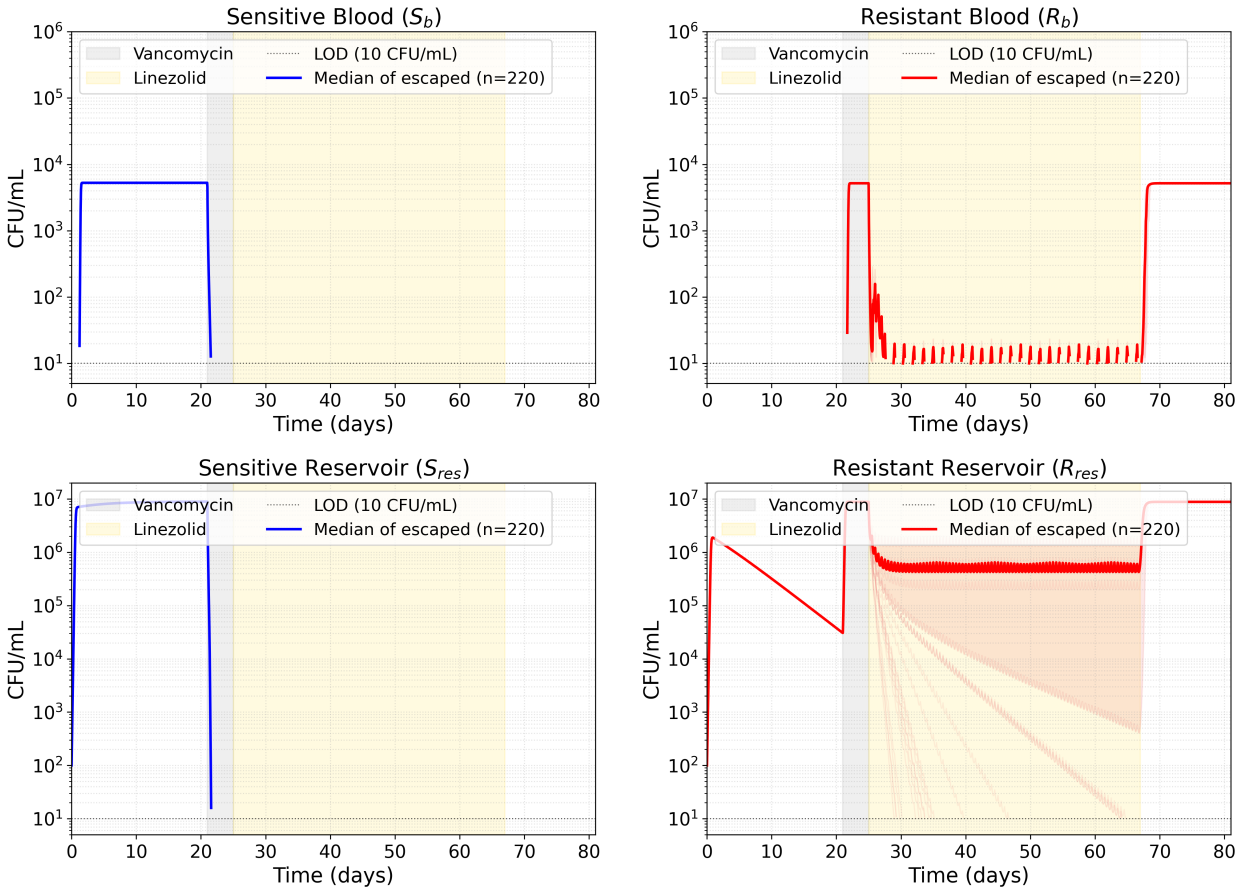


Fig. 2. Monte Carlo sweep of $EC_{50,L}$ ($n = 1000$): full time-course kinetics for S_b, R_b, S_{res}, and R_{res}, shown as individual trajectories (light), median (solid), and 5th–95th percentile envelope (shaded).

Increasing $EC_{50,L}$ represents decreasing linezolid potency and a greater probability of a lack of bacteremia resolution. The relapse of the bacteremia occurred in 22% of simulated cases, There was persistent bacteremia throughout treatment in 0.2% (two patients). This was the only result were bacteremia did not resolve during linezolid treatment. Complete resolution occurred in 78% of patient simulations. In fig. 3 the upper panel shows the bacterial load for the two compartments at the end of the follow up period of each simulation, given the value of $EC_{50,L}$. The lower panel illustrates the distribution of $EC_{50,L}$ values from the given distribution and their respective clinical result. The $EC_{50,L}$ calculated transition point between clinical resolution from failure is reported as $EC_{50,L} = 0.76$ mg/L. As illustrated in fig. 3 the majority of $EC_{50,L}$ probability mass is localized to the side of the transition boundary that favors clinical resolution.

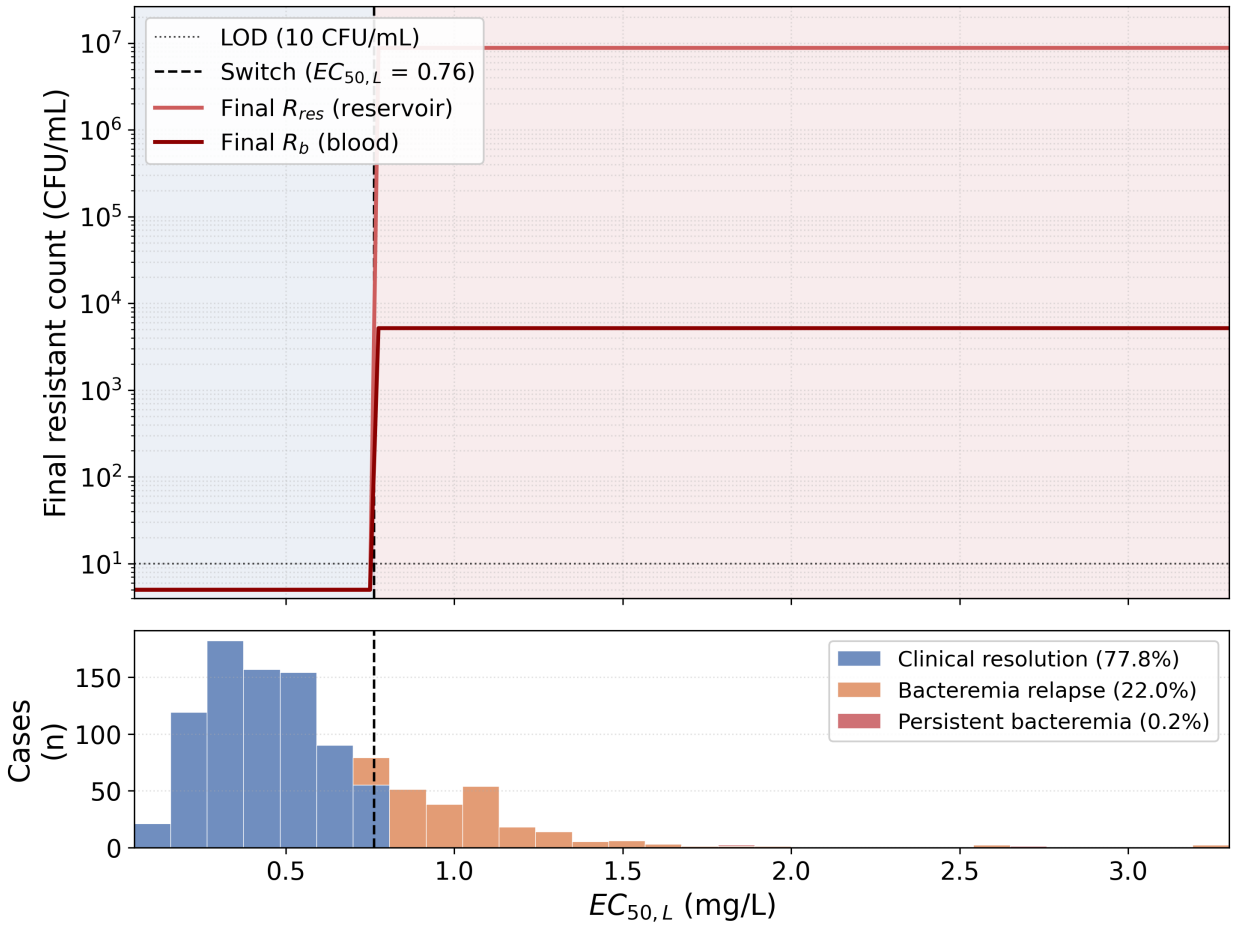


Fig. 3. Upper panel: Final bacterial burden in each compartment, Lower panel: Distribution of $EC_{50,L}$ values color coded with final clinical outcome.

The methodology for the calculation of the transition point is detailed in Section II-J.

D. Linezolid Maximum Effect

To assess how variation in linezolid maximum killing capacity affects predicted *Staph aureus* bacterial kinetics, $E_{max,L}$ was sampled from a mean-corrected log-normal distribution with arithmetic mean $1.97\ \mathrm{h}^{-1}$ and coefficient of variation $\mathrm{CV} = 0.080$. Published $E_{max,L}$ linezolid estimates span two distinct ranges: 0.735–$0.754\ \mathrm{h}^{-1}$ in Sandberg et al. [17] and 1.77–$2.22\ \mathrm{h}^{-1}$ in Scheerans et al. [35]. The present analysis uses the in vitro estimate of Vera-Yunca et al. ($E_{max,L} = 1.97\ \mathrm{h}^{-1}$, $EC_{50,L} = 0.56\ \mathrm{mg/L}$) [11]. The $E_{MAX,L}$ was scaled by 0.45, a value slightly higher than used by Vera-Yunca (0.303). Each of $n = 1000$ independently sampled $E_{max,L}$ values was propagated through the model over the complete 81-day treatment and follow-

up course fig. 4. There was eradication of bacteria from the blood compartment, however the bacterial load in the reservoir remained above the level of detection at the end of linezolid treatment.

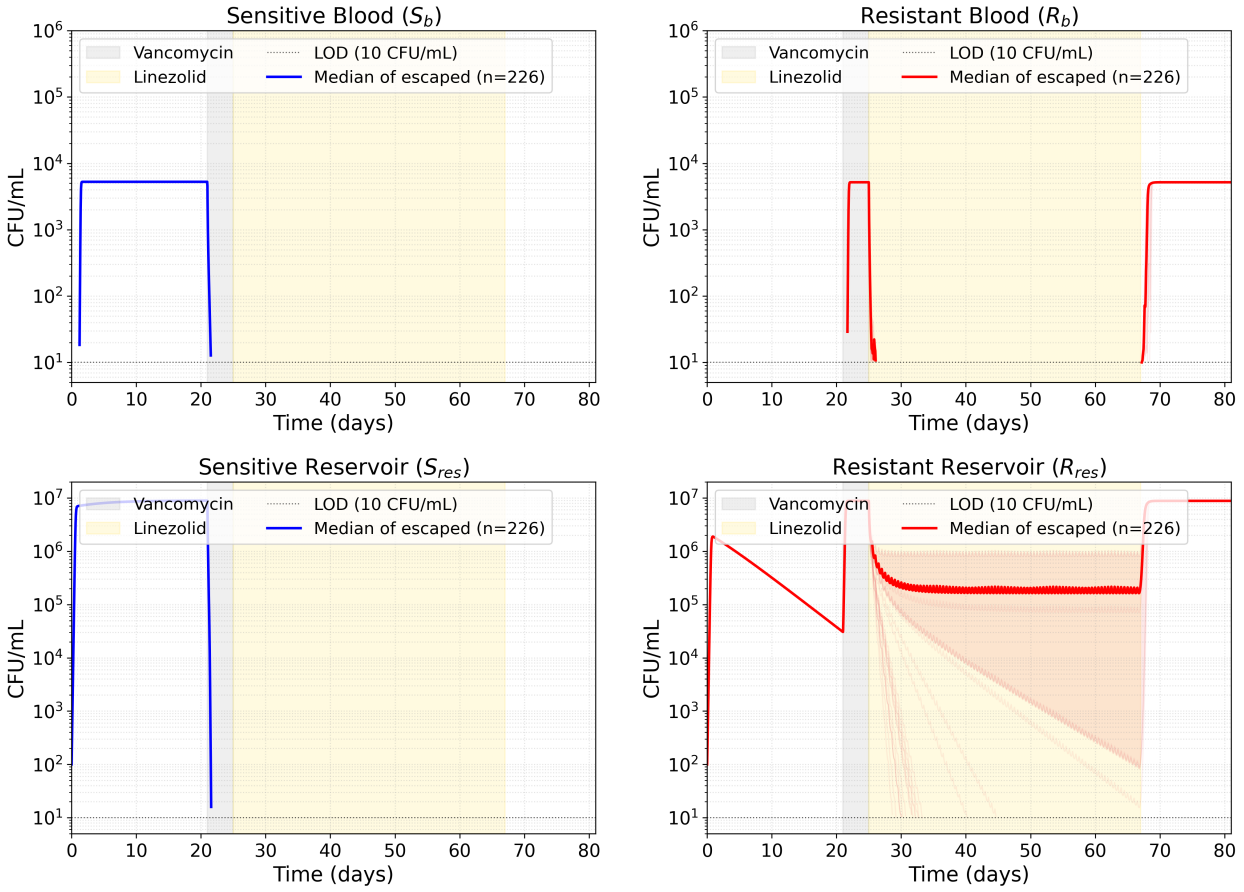


Fig. 4. Monte Carlo sweep of $E_{\max,L}$: S_b, R_b, S_{res} and R_{res}, with individual relapsed trajectories (light), the relapsed-patient median (solid), and relapsed-patient 5th–95th percentile envelope (shaded).

Lower $E_{max,L}$ values represent weaker antibiotic effect and poor response to treatment. Bacteremia resolution occurred in 77% of simulated patients, and a reseeding of the blood compartment from the reservoir in 23% of simulated patients. There no instances of persistent bacteremia during treatment fig. 5. The upper panel of fig. 5 shows the bacterial load for each compartment at the termination of each simulation. The lower panel of fig. 5 shows the distribution of $E_{max,L}$ values and their respective clinical result. The calculated transition point between treatment response and failure was reported at $1.85\ \text{h}^{-1}$. Most of the $E_{max,L}$ probability mass is located above the $1.85\ \text{h}^{-1}$ transition point, favoring clinical resolution.

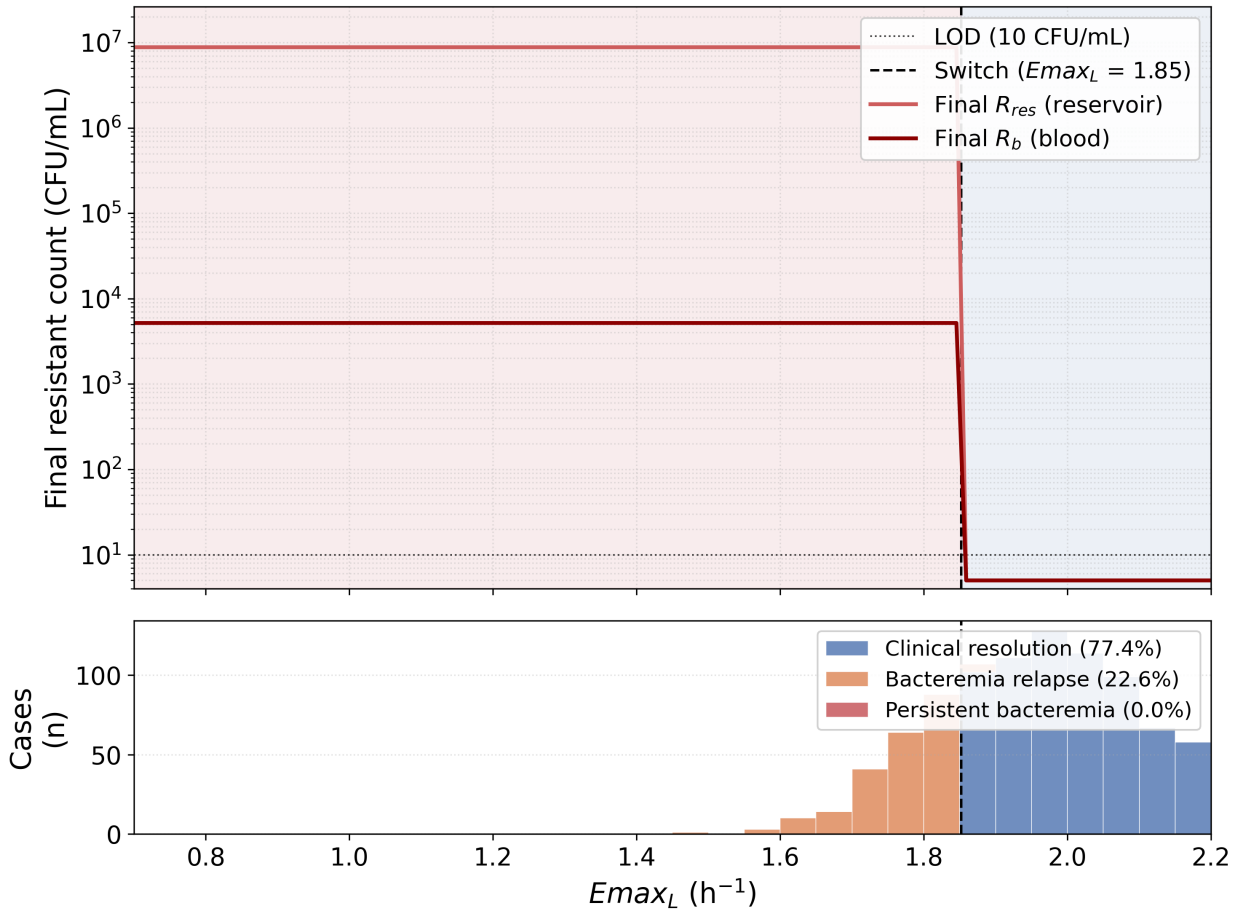


Fig. 5. Upper panel: Final bacterial burden in each compartment, Lower panel: Distribution of $E_{50,L}$ values color coded with final clinical outcome.

Methodology for transition point calculation is described in Section II-J.

E. Host Immune Response

The immune response was modeled with a constant linear function (k_{immune}) with a value of $0.18\ \mathrm{h}^{-1}$, which served as the mean of a log-normal distribution. A coefficient of variation $\mathrm{CV} = 0.80$ was used to provide a measure of biological variability. Although unlikely to be a true representation of a highly complex system, it does provide a simple method to incorporate the innate immune response into this model. The value of k_{immune} was arrived at by using the population average outcome for this specific clinical scenario and back calculating what value would be needed to obtain the expected clinical outcome based on population averages of 20-30%. This value is a calibration result and is not an independent prediction. All remaining

parameters were held at baseline. For each of $n = 1000$ independently sampled k_{immune} values were propagated through the dual-reservoir model over the complete 81-day treatment and follow-up course. A distribution of trajectories was created for each of the four bacterial subpopulations fig. 6.

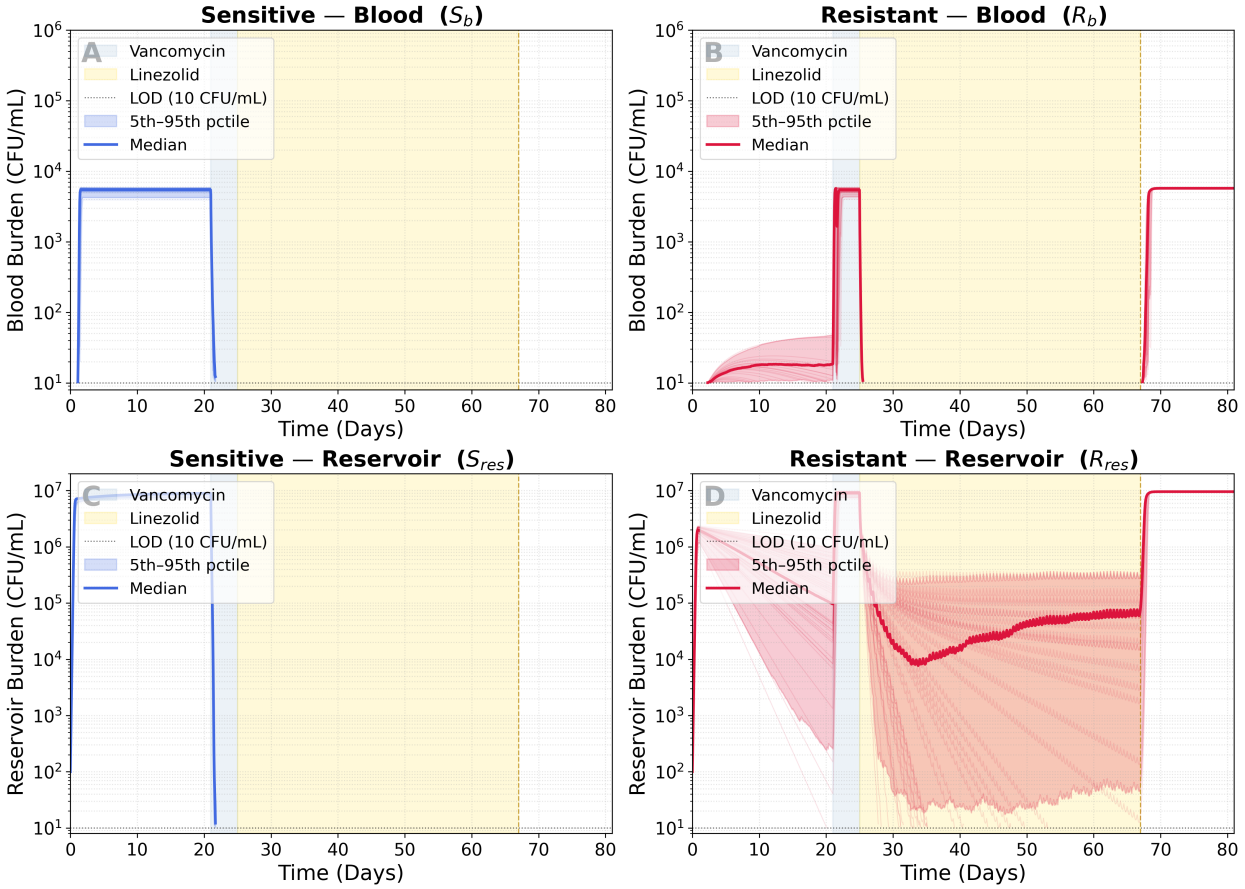


Fig. 6. Monte Carlo sweep of k_{immune} ($n = 1000$): full time-course kinetics for S_b, R_b, S_{res}, and R_{res}, shown as individual trajectories (light), median (solid), and 5th–95th percentile envelope (shaded).

Higher values of k_{immune} bias the outcomes toward clinical resolution of the bacteremia. Bacteremia resolution occurred in 72% of simulated patients, relapse in 28% and no persistent bacteremia. The transition point between clinical resolution from failure is reported as $k_{immune} = 0.093\ \mathrm{h}^{-1}$ fig. 7. The upper panel shows the bacterial load for each compartment at the termination of each simulation, given the value of k_{immune}. The lower panel illustrates the distribution of k_{immune} values from the given distribution.

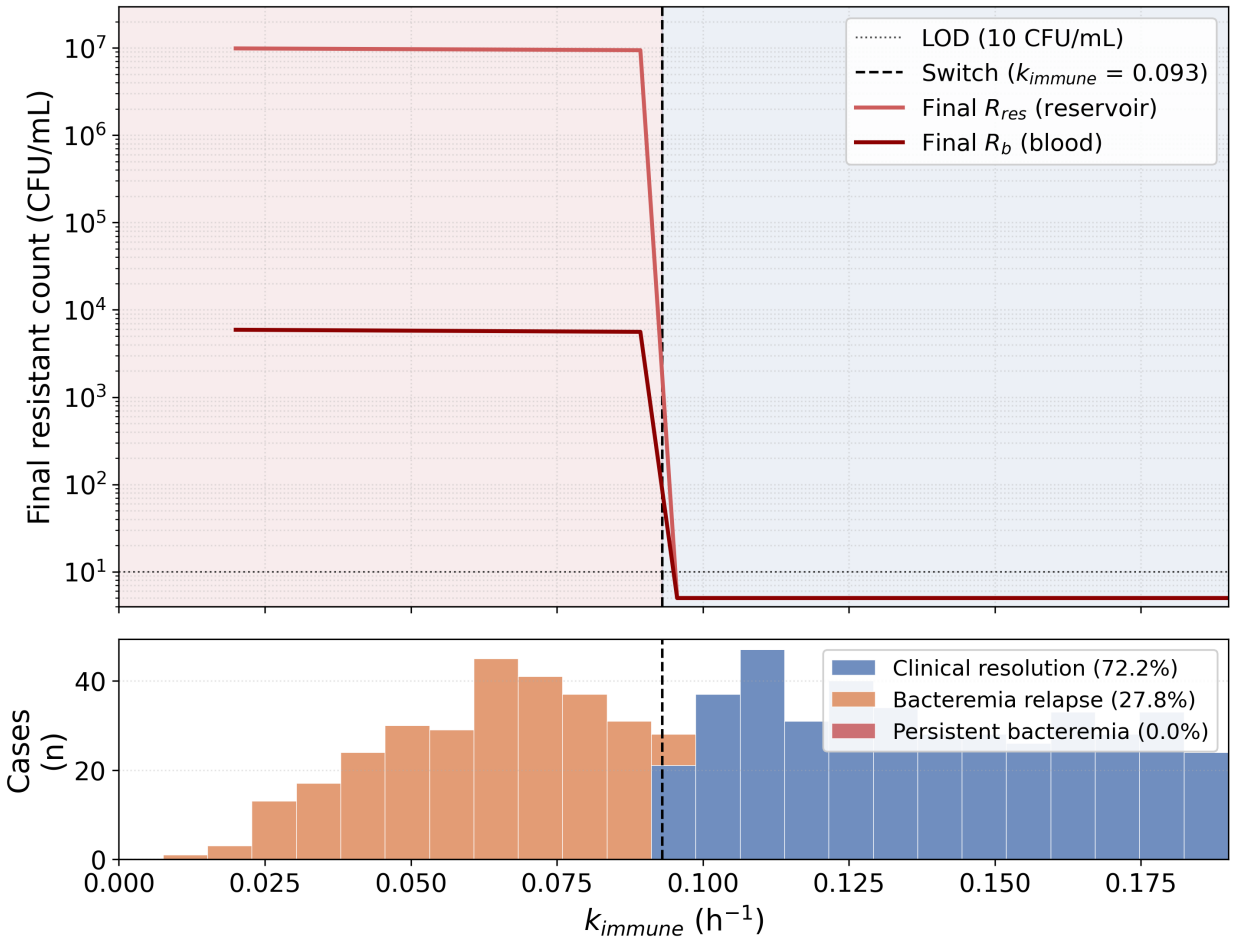


Fig. 7. Upper panel: Final bacterial burden each compartment, Lower panel: Distribution of k_{immune} values, color coded for final outcome.

F. Theoretical Clinical Response

Each parameter E_{max}, $EC_{50,L}$ and k_{immune} has a transition boundary separating the resolution of the bacteremia at the end of simulation from the presence of bacteremia at the end. The transition point $EC_{50,L}$ is 0.76 mg/L, E_{max} is $1.85\ \mathrm{h}^{-1}$ and immune clearance is $0.093\ \mathrm{h}^{-1}$. Using this information and the mean of each distribution the share of patients that clear the bacteremia can be calculated directly. The z-value is a measurement of how many standard deviations the transition boundary lies from the center of the distribution (on a log scale). The normal distribution converts the z value into the expected share of patients located on one side of the transition point.

Equations (15)– (16), evaluated at the transition points, gave

$$\begin{aligned} z_{EC_{50}} &= 0.83, \quad & p_{persist} &= 1 - \Phi(z_{EC_{50}}) = 20.2\%, \\ z_{immune} &= -0.59, \quad & p_{persist} &= \Phi(z_{immune}) = 27.9\%, \\ z_{E_{max}} &= -0.74, \quad & p_{persist} &= \Phi(z_{E_{max}}) = 23.1\%. \end{aligned} \tag{17}$$

For each parameter, the standardized log-space distance between the transition point and the center of the sampling distribution, $z_T = (\ln T - \mu_{\ln})/\sigma_{\ln}$, is converted through the standard normal cumulative distribution function into the expected fraction of patients on the no bacteremia resolution side of the transition, giving predicted fractions of 20.2% for $EC_{50,L}$ ($z = 0.83$), 23.1% for $E_{max,L}$ ($z = -0.74$), and 27.9% for k_{immune} ($z = -0.59$). These predictions agreed closely with the 1,000-patient Monte Carlo cohorts, in which 222 $EC_{50,L}$ samples (22.2%), 226 $E_{max,L}$ samples (22.6%), and 278 k_{immune} samples (27.8%) fell on the no bacteremia resolution side. The largest discrepancy, 2.0 percentage points for $EC_{50,L}$, is within the finite-sample variation expected for a cohort of this size (standard error about 1.3 percentage points).

Parameter	Mean	CV	Transition	Relapse	Persistent / Clinical Resolution
$EC_{50,L}$	0.56	0.60	0.76	22.0%	0.2% / 77.8%
$E_{max,L}$	1.97	0.080	1.85	22.6%	0.0% / 77.4%
k_{immune}	0.18	0.80	0.093	27.8%	0.0% / 72.2%

TABLE IV

UNIFIED SAMPLING INPUTS, RESERVOIR-PERSISTENCE TRANSITIONS, AND VERIFIED MONTE CARLO OUTCOMES. UNITS ARE MG/L FOR $EC_{50,L}$ AND H^{-1} FOR $E_{max,L}$ AND k_{immune}.

Table IV provides the cross-parameter comparison of the parameter distributions (mean and CV), the transition boundaries, and simulated Monte Carlo outcomes.

Figure 8 visualizes the distributional calculation behind these outcomes. Although the parameters have different units and coefficients of variation, each calibration places approximately one-fifth to three-tenths of its probability mass on the no bacteremia resolution side of the transition point. The theoretical tail probabilities and realized 1,000-patient fractions are shown together so that finite-sample variation is visible.

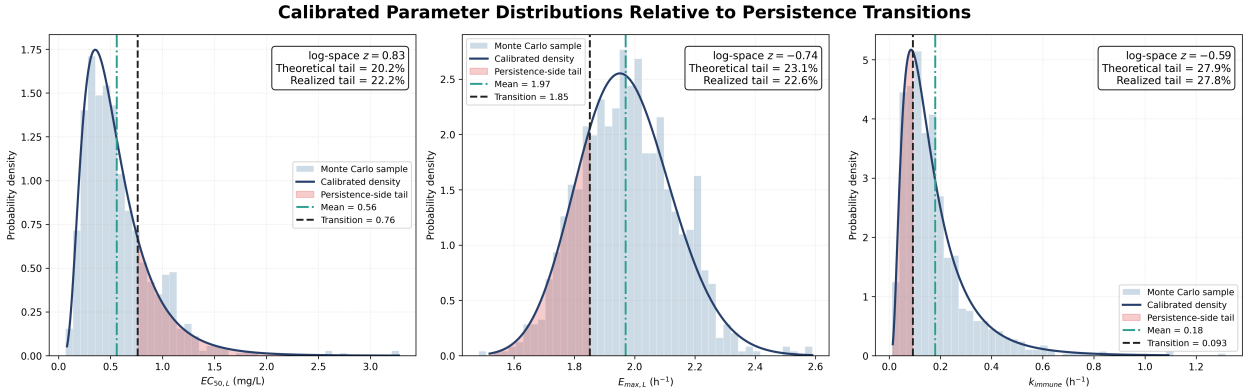


Fig. 8. Parameter distributions relative to their transition points. Histograms show the realized Monte Carlo samples, solid curves show the probability densities, and red shading identifies the persistence-side tail. The panels report the log-space z value and corresponding normal-CDF tail probability for each mean-corrected log-normal distribution.

The cumulative incidence curves in Figure 9 show that the return of the bacteremia in the blood compartment during the 14-day follow-up occur early in the post treatment phase of the simulation. All three curves reach their final cohort relapse percentages within approximately two days after linezolid completion. The denominator is the full 1,000-patient cohort for each curve.

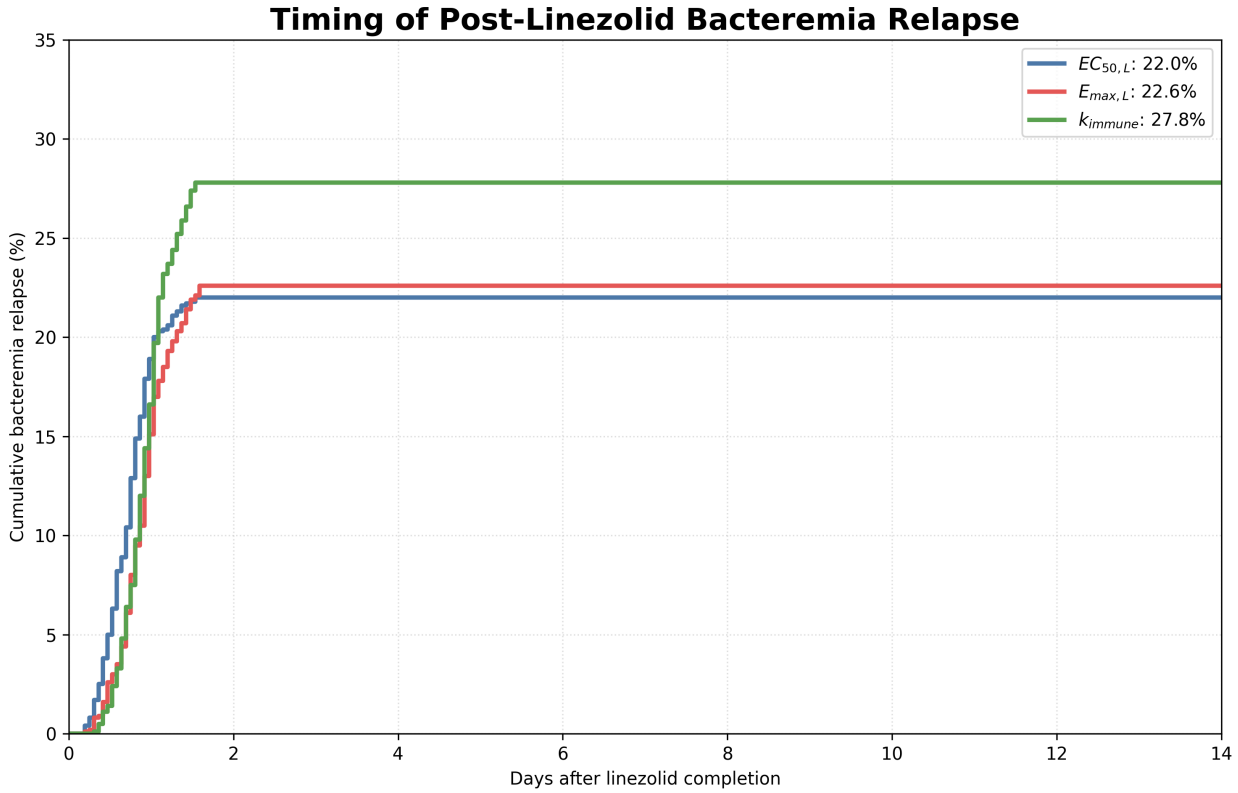


Fig. 9. Cumulative incidence and timing of bacteremia relapse after linezolid completion. Each step marks the first post-treatment return of resistant blood bacteria to the 10 CFU/mL LOD. Persistent treatment failures are excluded because their blood burden was already detectable when linezolid ended.

IV. Discussion

This goal of our study was to construct a mathematical model to represent the development of bacterial endocarditis in a human patient. A dual-compartment PKPD framework was used in order to incorporate the key components of the infection. Osteomyelitis was represented by the reservoir, hematogenous spread of the infection from the reservoir to the mitral valve was represented by the blood compartment. The model pathogen was MRSA, composed of two sub-populations for sensitive and resistant vancomycin. Treatment followed standard recommended protocols from IDSA and used vancomycin as first line treatment, followed by linezolid once the presence of vancomycin resistance was detected. The presence of endocarditis was confirmed by TEE. Due to the lack of human data needed to construct a PKPD model other sources were used to help guide model construction. Where human data were available they were used to inform model construction.

Multiple studies were reviewed for suitability, including Schmidt, Scheerans, Vera-Yunca and Mikkaichi. The study by Vera-Yunca et al. was selected to provide the values for several of the key parameters of our model. Although not an exact match to our model system it was the most similar compared to the other reviewed studies. Our mean values for bacterial growth of both subpopulations, $EC_{50,L}$, $E_{max,L}$ and reservoir carry capacity were taken from their study. Resistant MRSA growth was assigned a fitness cost of 10%. The distribution of each parameter was modeled with the log-normal distribution based on the mean values from Vera-Yunca. A scaling factor was used to lower the $E_{max,L}$ between the blood and reservoir compartments. Our scaling factor was slightly larger than the value used by Vera-Yunca.

The importance of the immune system in resolving infections of this type cannot be overestimated. Our current understanding of the key factors of the immune system that are responsible for clearing an infection is limited and incomplete. In view of this reality and in order to minimize computational complexity a simple linear term was used to represent the immune system. The effect of the immune response in the reservoir was scaled 50% of immune response in the blood compartment. The value of the linear term was a calibrated based on the constraints imposed by the other parameters of our system.

Multiple prior human studies have reported relapse rates between 20 to 30 percent for lone bone osteomyelitis, represented by the reservoir compartment. Using these estimates, the parameter estimates from Vera-Yunca, and other studies the remaining parameter values were estimated. These values included the linear term for the immune response, k_{immune}, the $E_{max,L}$ scaling factor. The PK parameters, such as V_d, clearance rate and antibiotic bone penetration were based on population averages commonly used in clinical practice. Normal kidney function was assumed. Bacteremia relapse was defined in this context as the presence of bacteria above the LOD at the end of the follow-up period.

Brent's method was used to calculate the transition points in the system that separate clinical resolution from failure. These transition points mark a change in the outcome of the simulation at the end of follow-up. They resemble bifurcation points in that a small change in a parameter produces a qualitatively different outcome, but they have not been shown to be bifurcations of the underlying system: no analysis of its equilibria or their stability was performed, and the transition also depends on the finite treatment course, the finite observation time, and the extinction rule at the LOD. The distance between the transition point and distributional means were constructed in terms of z values. Theoretical probabilities for clinical resolution and failure were constructed

for each of the 3 parameters. The $E_{max,L}$ analysis produced 226 theoretical clinical failures (22.6%), 0 theoretical persistent failures (0.0%), and 774 theoretical bacteremia resolutions (77.4%), with a transition point at $1.85\ \mathrm{h}^{-1}$. The immune analysis produced 278 theoretical clinical failures (27.8%), 0 theoretical persistent failures (0.0%), and 722 theoretical bacteremia resolutions (72.2%), with a transition point at $0.093\ \mathrm{h}^{-1}$. The $EC_{50,L}$ analysis produced 220 theoretical clinical failures (22.0%), 2 theoretical persistent treatment failures (0.2%), and 778 theoretical bacteremia resolutions (77.8%), with a transition point at $0.76\ \mathrm{mg/L}$.

Given our model structure 1000 Monte Carlo simulations were performed and results were compared to the transition point estimates. The results of the $EC_{50,L}$ Monte Carlo simulation returned clinical failure in 22.2% of the cohort (22.0% relapse and 0.2% persistent bacteremia) and 77.8% clinical resolution of the bacteremia. Similar results can be seen for $E_{max,L}$ and k_{immune}. The Monte Carlo results closely agree with the theoretical estimates based on transition points.

The probability of bacteremia resolution (or relapse) can be thought of in terms of the size and distributional structure of the probability mass in relation with the transition points of the system. Moving the $EC_{50,L}$ distribution upward or increasing variability so that more mass enters the distributional tail would increase the fraction of the cohort that fails to clear the bacteremia. The opposite is true for the $E_{max,L}$ or k_{immune}. Moving the distributions downward below the respective transition points would lead to failure to clear the bacteremia. While it is recognized that the transition points of the system are based on the the mean values chosen for the distributions and cannot be considered independent they do provide a way to think about what factors affect clinical response to treatment. These results were comparable to the Monte Carlo results.

This study provides a mathematical framework that allows one to better understand how different variables affect individual patient outcomes to antibiotic treatment in the bacterial endocarditis setting. Our study examined multiple complex issues including bacterial resistance, biofilms, a complex disease process where there are two infections. The scenario is based on a true patient and illustrates the type of complex clinical scenario commonly seen in clinical practice. It continues to be a challenge to treat successfully despite the progress in antibiotic development over the past 30 years. Current technological limitations in collecting real time data from live patients makes simulation our best tool in studying this type of complex clinical problem. It is our desire that this study will provide a starting point to refine and further develop

this approach. The python code is available for download from our GitHub site for further investigation.

V. Limitations and Future Directions

This simulation study carries several limitations driven by current gaps in empirical data. The model relies on several assumptions concerning the bone reservoir compartment. Key parameters such as inter-compartmental transfer rates, intra-bone bacterial growth and death rates, and the baseline densities of sensitive and resistant populations are not directly measured and were estimated. Transfer between bone and blood was modeled as one-way, from bone into blood only; re-seeding of the bone focus from the circulation was neglected. While this simplification is biologically plausible (an established focus sheds into the bloodstream far more than the circulation seeds it), the reservoir-to-blood rate itself is an estimated value. None of the values used in this part of the analysis were based on data.

Second, the host immune response is represented by a single first-order clearance rate, k_{immune}, applied uniformly to each compartment (with a lower effective rate in bone than in blood, reflecting reduced immune access to a deep-seated focus). This rate had to be estimated, and the model does not attempt to represent the underlying cellular or molecular mechanisms of clearance. Neutrophil recruitment, killing dynamics, the networks of cytokines[1] and chemokines[2] that coordinate them, are important in guding the innate immune response and are omitted in favor of this single linear aggregate rate. It is highly unlikely that immune function can be modeld by a linear function.

Third, the blood carrying capacity for *S. aureus* is based on a limited set of measurements using PCR-based quantification of bacterial DNA, and the maximal carrying capacity used in the model was derived from this same limited dataset.

Fourth, published estimates of EC_{50} do not consistently distinguish between total drug concentration and the free (unbound) fraction in plasma; this model used free serum concentration throughout its pharmacokinetic component, which may not match the basis of every EC_{50} value drawn from the literature.

Fifth, the resistant strain carries a fitness cost in both compartments — 10% in both blood ($\rho_r = 1.323$ versus $\rho_s = 1.47\ \mathrm{h}^{-1}$) and the reservoir ($\rho_r^r = 0.761$ versus $\rho_s^r = 0.845\ \mathrm{h}^{-1}$) — but

[1]Signaling molecules that regulate inflammation, cell growth, and immune cell activity.

[2]A family of small proteins that attract and guide immune cells to specific sites of infection.

the cost is assumed rather than measured. The sensitive growth rates are taken from a single study [11]: the in vitro rate for blood and the in vivo mouse-lung rate for bone, neither of which was measured in bone. The reservoir growth rates should therefore be read as calibrated quantities chosen to reproduce the clinical scenario, not as independently measured biological values. The reservoir linezolid efficacy factor ($\alpha = 0.45$) is likewise a calibrated quantity rather than a measured one, and the outcomes are sensitive to it: small changes in α shift the relapse fraction of every analysis substantially.

A further limitation is that the unified fixed baselines and Monte Carlo means ($EC_{50,L} = 0.56$ mg/L, $E_{max,L} = 1.97$ h^{-1}, and $k_{immune} = 0.18$ h^{-1}) have not been externally validated as a joint parameter set. Any Monte Carlo percentages are conditional on this parameterization and the specified distributions. The coefficients of variation (0.60 for $EC_{50,L}$, 0.080 for $E_{max,L}$, and 0.80 for k_{immune}) are outcome-calibrated model inputs, not independently estimated population-variability parameters. External calibration and validation against measured blood and bone burdens are required before these outputs can be interpreted as patient risk estimates.

The extinction boundary at the LOD is also a modeling assumption. Bacteria transferred from the reservoir into a blood subpopulation that has fallen below the LOD are accumulated without loss: until the accumulated amount reaches the LOD, it is not subject to antibiotic killing or immune clearance, and it is retained even if the reservoir subpopulation that supplied it later becomes extinct. In a patient, some of these organisms would be removed by linezolid or by the immune response before they could re-establish a bacteremia. This rule therefore favors reseeding of the blood compartment and may overestimate the frequency of relapse.

Finally, the in vitro bacterial doubling times used for calibration may not fully reflect the metabolically dormant or small-colony-variant (SCV) states typical of chronic biofilm infections, nor do they necessarily reflect the actual replication rate of MRSA within the circulatory system in vivo. All growth-rate calibration in this model is ultimately anchored to in vitro measurements, and in vivo replication dynamics – particularly within a biofilm or vegetative focus – remain a source of uncertainty this model cannot resolve.

Additional limitations stem from constraints on available patient-level data and the scope of the sensitivity analyses performed. Because patient-level data on model parameters such as growth rates of MRSA in blood pharmacodynamic and other physiological variables remain limited, the model could not be formally validated against clinical outcome data. The Monte Carlo sensitivity analyses were also univariate: each sweep varied a single parameter ($EC_{50,L}$, $E_{max,L}$,

or k_{immune}) while holding every other parameter fixed at baseline. Real inter-patient variability is likely joint and correlated. Impaired renal function, can simultaneously reduce drug clearance and also alter immune status. This univariate approach cannot capture these interaction effects between parameters that vary together in an actual patient population. Similarly, no attempt was made to introduce inter-patient pharmacokinetic variability: volume of distribution, clearance, and protein binding were held fixed at population-average values for every simulated patient. Finally, the simulation used a single fixed treatment protocol throughout, with no exploration of missed doses, alternative dosing regimens, or source control (e.g. surgical debridement or hardware removal), all of which materially affect outcomes in real bone and endocarditis infections.

REFERENCES

[1] M. Li, J. B. Kim, B. K. S. Sastry, and M. Chen, "Infective endocarditis," *The Lancet*, vol. 404, pp. 377–392, 2024. Seminar.

[2] M. Hammond-Haley, A. Hartley, B. M. Al-Khayatt, A. J. Delago, A. Ghajar, U. Ojha, D. C. Marshall, J. D. Salciccioli, B. D. Prendergast, and J. Shalhoub, "Trends in the incidence and mortality of infective endocarditis in high-income countries between 1990 and 2019," *International Journal of Cardiology*, vol. 371, pp. 441–451, 2023.

[3] S. S. Taduru, "30-year trends of incidence and mortality of infective endocarditis in the united states—unveiling the age- and gender-related and regional disparities," *The American Journal of Cardiology*, vol. 203, pp. 421–422, 2023.

[4] D. R. Murdoch, G. R. Corey, B. Hoen, J. M. Miró, V. G. Fowler, A. S. Bayer, A. W. Karchmer, L. Olaison, P. A. Pappas, P. Moreillon, S. T. Chambers, V. H. Chu, V. Falcó, D. J. Holland, P. Jones, J. L. Klein, N. J. Raymond, K. M. Read, M. F. Tripodi, R. Utili, A. Wang, C. W. Woods, and C. H. Cabell, "Clinical presentation, etiology, and outcome of infective endocarditis in the 21st century: The international collaboration on endocarditis–prospective cohort study," *Archives of Internal Medicine*, vol. 169, no. 5, pp. 463–473, 2009.

[5] D. J. Minter, A. Appa, H. F. Chambers, and S. B. Doernberg, "Contemporary management of *Staphylococcus aureus* bacteremia—controversies in clinical practice," *Clinical Infectious Diseases*, vol. 77, no. 11, pp. e57–e68, 2023.

[6] B. E. Murray, C. A. Arias, and E. C. Nannini, "Glycopeptides (vancomycin and teicoplanin) and lipoglycopeptides (telavancin, oritavancin, and dalbavancin)," in *Mandell, Douglas, and Bennett's Principles and Practice of Infectious Diseases*, ch. 30, pp. 375–395.e4, Philadelphia: Elsevier, 9 ed., 2019.

[7] W. A. McGuinness, N. Małachowa, and F. R. DeLeo, "Vancomycin resistance in staphylococcus aureus," *The Yale Journal of Biology and Medicine*, vol. 90, no. 2, pp. 269–281, 2017.

[8] B. C. Kahl, K. Becker, and B. Löffler, "Clinical significance and pathogenesis of staphylococcal small colony variants in persistent infections," *Clinical Microbiology Reviews*, vol. 29, pp. 401–427, apr 2016.

[9] Y. Cong, S. Yang, and X. Rao, "Vancomycin-resistant *Staphylococcus aureus* infections: a review of case updating and clinical features," *Journal of Advanced Research*, vol. 21, pp. 169–176, 2020.

[10] T. Mikkaichi, M. R. Yeaman, A. Hoffmann, and MRSA Systems Immunobiology Group, "Identifying determinants of persistent MRSA bacteremia using mathematical modeling," *PLOS Computational Biology*, vol. 15, p. e1007087, July 2019.

[11] D. Vera-Yunca, C. Matias, C. V. Lundberg, and L. E. Friberg, "Model-based translation of the pkpd-relationship for linezolid and vancomycin on methicillin-resistant *Staphylococcus aureus*: from in vitro time–kill experiments to a mouse pneumonia model," *Journal of Antimicrobial Chemotherapy*, vol. 80, no. 7, pp. 1860–1868, 2025.

[12] P. Virtanen, R. Gommers, T. E. Oliphant, M. Haberland, T. Reddy, D. Cournapeau, E. Burovski, P. Peterson, W. Weckesser, J. Bright, S. J. van der Walt, M. Brett, J. Wilson, K. J. Millman, N. Mayorov, A. R. J. Nelson, E. Jones, R. Kern, E. Larson, C. J. Carey, İ. Polat, Y. Feng, A. W. Moore, J. VanderPlas, D. Laxalde, J. Perktold, R. Cimrman, I. Henriksen, E. A. Quintero, C. R. Harris, A. M. Archibald, A. H. Ribeiro, F. Pedregosa, P. van Mulbregt, and SciPy 1.0 Contributors, "SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python," *Nature Methods*, vol. 17, pp. 261–272, 2020.

[13] E. Garcia del Pozo, J. Collazos, J. A. Carton, D. Camporro, and V. Asensi, "Factors predictive of relapse in adult bacterial osteomyelitis of long bones," *BMC Infectious Diseases*, vol. 18, no. 1, p. 635, 2018.

[14] A.-E. Nieman, W. Rozemeijer, P. H. M. Savelkoul, and R. P. Schade, "Bacterial dna load in staphylococcus aureus bacteremia is significantly higher in intravascular infections," *PLoS ONE*, vol. 17, no. 4, p. e0266869, 2022.

[15] I. I. Momodu and V. Savaliya, "Osteomyelitis," in *StatPearls*, Treasure Island, FL: StatPearls Publishing, 2025. Updated May 31, 2023.

[16] M. Kostakioti, M. Hadjifrangiskou, and S. J. Hultgren, "Bacterial biofilms: Development, dispersal, and therapeutic strategies in the dawn of the postantibiotic era," *Cold Spring Harbor Perspectives in Medicine*, vol. 3, no. 4, p. a010306, 2013.

[17] A. Sandberg, K. S. Jensen, P. Baudoux, F. Van Bambeke, P. M. Tulkens, and N. Frimodt-Moller, "Intra- and extracellular activity of linezolid against Staphylococcus aureus in vivo and in vitro," *Journal of Antimicrobial Chemotherapy*, vol. 65, no. 5, pp. 962–973, 2010.

[18] N. Feweks, "Vanco PK review." https://vancopk.com/vanco-kinetics-review/, 2025. Accessed 29 November 2025.

[19] G. R. Matzke, G. G. Zhanel, and D. R. P. Guay, "Clinical pharmacokinetics of vancomycin," *Clinical Pharmacokinetics*, vol. 11, no. 4, pp. 257–282, 1986.

[20] M. S. Dryden, "Linezolid pharmacokinetics and pharmacodynamics in clinical treatment," *Journal of Antimicrobial Chemotherapy*, vol. 66, no. Suppl 4, pp. iv7–iv15, 2011.

[21] F. Kutscha-Lissberg, U. Hebler, G. Muhr, and M. Köller, "Linezolid penetration into bone and joint tissues infected with methicillin-resistant staphylococci," *Antimicrobial Agents and Chemotherapy*, vol. 47, no. 12, pp. 3964–3966, 2003.

[22] B. Spellberg and B. A. Lipsky, "Systemic antibiotic therapy for chronic osteomyelitis in adults," *Clinical Infectious Diseases*, vol. 54, no. 3, pp. 393–407, 2012.

[23] M. J. Rybak, J. Le, T. P. Lodise, D. P. Levine, J. S. Bradley, C. Liu, B. A. Mueller, M. P. Pai, A. Wong-Beringer, J. C. Rotschafer, K. A. Rodvold, H. D. Maples, and B. M. Lomaestro, "Therapeutic monitoring of vancomycin for serious methicillin-resistant staphylococcus aureus infections: A revised consensus guideline and review by the american society of health-system pharmacists, the infectious diseases society of america, the pediatric infectious diseases society, and the society of infectious diseases pharmacists," *American Journal of Health-System Pharmacy*, vol. 77, no. 11, pp. 835–864, 2020.

[24] A. Alsultan, "Determining therapeutic trough ranges for linezolid," *Saudi Pharmaceutical Journal*, vol. 27, pp. 1061–1063, 2019.

[25] D. Mollison, "Dependence of epidemic and population velocities on basic parameters," *Mathematical Biosciences*, vol. 107, no. 2, pp. 255–287, 1991.

[26] F. Campillo and C. Lobry, "Effect of population size in a predator–prey model," *Ecological Modelling*, vol. 246, pp. 1–10, 2012.

[27] SciPy Development Team, *SciPy Reference Guide*. Enthought, Inc. and SciPy Community, 2024. Accessed: 2026-03-25.

[28] Wolfram Research, "Events and discontinuities in differential equations." Wolfram Language Documentation. Accessed: 2026-09-29.

[29] W. H. Enright, K. R. Jackson, S. P. Nørsett, and P. G. Thomsen, "Effective solution of discontinuous ivps using a runge–kutta formula pair with interpolants," *Applied Mathematics and Computation*, vol. 27, no. 4, pp. 313–335, 1988.

[30] L. F. Shampine and S. Thompson, "Event location for ordinary differential equations," *Computers & Mathematics with Applications*, vol. 39, pp. 43–54, 2000.

[31] R. P. Brent, "An algorithm with guaranteed convergence for finding a zero of a function," *The Computer Journal*, vol. 14, no. 4, pp. 422–425, 1971.

[32] R. P. Brent, *Algorithms for Minimization without Derivatives*. Englewood Cliffs, New Jersey: Prentice-Hall, 1973.

[33] SciPy Developers, "scipy.optimize.brentq — SciPy api reference." Accessed 4 September 2026.

[34] E. Limpert, W. A. Stahel, and M. Abbt, "Log-normal distributions across the sciences: Keys and clues," *BioScience*, vol. 51, pp. 341–352, May 2001.

[35] C. Scheerans, S. G. Wicha, J. Michael, H. Derendorf, and C. Kloft, "Concentration–response studies and modelling of the pharmacodynamics of linezolid: *Staphylococcus aureus* versus *Enterococcus faecium*," *International Journal of Antimicrobial Agents*, vol. 45, no. 1, pp. 54–60, 2015.